

In-Ground Silo Rebuild Station for OHT Wheel Groups

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

In-Ground Silo Rebuild Station for OHT Wheel
Groups..... 1

1.0 Introduction 2

2.0 Best Practice Description 2

3.0 Implementation Steps 4

4.0 Benefits 4

5.0 Resources Required 5

6.0 Supporting Attachments..... 5

7.0 Related Best Practices..... 6

8.0 Acknowledgements..... 6



1.0 Introduction

The size of large Off Highway Truck Wheel Groups makes rebuild difficult. Wheel group rebuild procedure requires the wheel group to be positioned vertically for disassembly and assembly. Technicians work with ladders and work platforms to rebuild the larger model Wheel Groups. A new wheel group rebuild work station concept consists of an in ground silo, a rotating platform to hold the wheel group and rotate the spindle for wheel bearing preload adjustment, and a hydraulic motor and adjustable lift to raise and lower the wheel station to any desired height. Technicians can work on wheel stations at waist height and eliminate the need for ladders and platforms for disassembly and assembly.

2.0 Best Practice Description

The common practice for Off Highway Truck wheel group rebuild is to stand the wheel station vertically on the shop floor for disassembly and assembly. Larger wheel groups are up to 12 feet tall. Disassembly and assembly requires technicians to work with ladders and work platforms in order to reach the internal parts of the wheel group.

The in ground silo concept positions the wheel station on a rotating platform mounted on a hydraulic lift which is installed in a silo or in the shop floor.

The lift can raise or lower the wheel group to any desired height, allowing mechanics to work on the wheel station at waist height without the need for ladders and platforms. The hydraulic motor also can rotate the wheel station spindle to seat and adjust the main bearing preload. A similar set-up without rotating capability can be used in the disassembly area for adjusting component height.



THE INFORMATION HEREON IS THE PROPERTY OF CATERPILLAR INC. AND/OR ITS SUBSIDIARIES. WITHOUT WRITTEN PERMISSION, ANY COPYING, TRANSMITTAL TO OTHERS, AND ANY USE EXCEPT THAT FOR WHICH IT IS LOANED, IS PROHIBITED.

In Ground Silo Rebuild Station for OHT Wheel Groups

DATE
6/7/2017

CHG
NO
00

NUMBER
0906-4.3-1012

Final Drive D/A Fixture:

- Positions and secures the spindle and wheel group
- Clamp and mounting system adjusts for 777-797 rear Wheel Station
- Rotates spindle 0-4 rpm for seating and setting bearing pre-load
- Hydraulic power unit and controls included
- 40,000 lb capacity lift system available for pit installation in facilities with restricted crane hook height

THE INFORMATION HEREON IS THE PROPERTY OF CATERPILLAR INC. AND/OR ITS SUBSIDIARIES. WITHOUT WRITTEN PERMISSION, ANY COPYING, TRANSMITTAL TO OTHERS, AND ANY USE EXCEPT THAT FOR WHICH IT IS LOANED, IS PROHIBITED.

In Ground Silo Rebuild Station for OHT Wheel Groups

DATE
6/7/2017

CHG
NO
00

NUMBER
0906-4.3-1012



3.0 Implementation Steps

1. A silo is constructed in the shop floor to contain the rotating platform, hydraulic motor, and lift. (See attached drawing for specifications)
2. Install hydraulic motor, platform, and lift. (See attachments for specifications)

Contact Caterpillar Service Tool Consultant prior to prep or purchase.

4.0 Benefits

- Significant improvement in labor efficiency, and overall less time to rebuild a wheel group.
- Improved rebuild precision in the installation of seals and other components.
- Improved precision in setting the main bearing preload.
- Improved technician comfort and work environment.
- Reduces overhead crane hook height requirement.
- Eliminates the need for platforms and steps.

THE INFORMATION HEREON IS THE PROPERTY OF CATERPILLAR INC. AND/OR ITS SUBSIDIARIES. WITHOUT WRITTEN PERMISSION, ANY COPYING, TRANSMITTAL TO OTHERS, AND ANY USE EXCEPT THAT FOR WHICH IT IS LOANED, IS PROHIBITED.

In Ground Silo Rebuild Station for OHT Wheel Groups

DATE
6/7/2017

CHG
NO
00

NUMBER
0906-4.3-1012

5.0 Resources Required

Investment includes

- Construction of a silo work-station in the shop floor,
- Installing a rotating wheel group platform with a hydraulic motor and lift.

Cost ranges from \$100,000 to \$150,000 depending on the size and capacity of the work-station. Also required, technician training in the station's proper use and safety.

6.0 Supporting Attachments

Tool Operating Manual NEHS1172

THE INFORMATION HEREON IS THE PROPERTY OF CATERPILLAR INC. AND/OR ITS SUBSIDIARIES. WITHOUT WRITTEN PERMISSION, ANY COPYING, TRANSMITTAL TO OTHERS, AND ANY USE EXCEPT THAT FOR WHICH IT IS LOANED, IS PROHIBITED.

In Ground Silo Rebuild Station for OHT Wheel Groups

DATE
6/7/2017

CHG
NO
00

NUMBER
0906-4.3-1012

